

Business Problem Statement

The Titanic disaster resulted in significant loss of life, but survival rates were not random. There were patterns related to passenger demographics, ticket class, gender, age, fare paid, and embarkation port.

Business Objective:

To analyze passenger data from the Titanic to:

- Identify key factors that influenced survival
- Understand demographic and socioeconomic disparities in survival rates
- Provide data-driven insights that explain survival patterns
- Build a dashboard to support exploratory and predictive analysis

Key Business Questions:

1. What was the overall survival rate?
2. Did gender influence survival probability?
3. Did passenger class (proxy for socioeconomic status) affect survival?
4. Were children more likely to survive?
5. Did fare amount correlate with survival?
6. Did embarkation port impact survival outcomes?

Key Performance Indicators (KPIs)

These KPIs should be displayed at the top of a dashboard.

Overall KPIs

1. **Total Passengers**
 2. **Total Survivors**
 3. **Total Non-Survivors**
 4. **Overall Survival Rate (%)**
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Demographic KPIs

5. **Survival Rate by Gender**
 - % Male survived

- % Female survived

6. Survival Rate by Age Group

- Child (0–12)
 - Teen (13–19)
 - Adult (20–59)
 - Senior (60+)
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Socioeconomic KPIs

7. Survival Rate by Passenger Class

- 1st Class
- 2nd Class
- 3rd Class

8. Average Fare Paid by Survivors vs Non-Survivors

Operational KPIs

9. Survival Rate by Embarkation Port

- Southampton
- Cherbourg
- Queenstown

10. Family Size Impact

- Survival rate by family size group
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Chart Requirements

Below are recommended charts for a business intelligence dashboard (Power BI / Tableau / Excel / Python).

1. Survival Distribution

Chart Type: Donut / Pie Chart

Purpose: Show proportion of survivors vs non-survivors

Metric: Count of Survived (0 vs 1)

2. Survival by Gender

Chart Type: Clustered Bar Chart

X-Axis: Gender

Y-Axis: Survival Rate (%)

Purpose: Compare survival likelihood between males and females

3. Survival by Passenger Class

Chart Type: Stacked Column Chart

X-Axis: Passenger Class

Legend: Survived vs Not Survived

Purpose: Show survival distribution across classes

4. Age Distribution by Survival

Chart Type: Histogram or Box Plot

X-Axis: Age

Color/Legend: Survived vs Not Survived

Purpose: Identify age patterns in survival

5. Fare vs Survival

Chart Type: Box Plot

X-Axis: Survival

Y-Axis: Fare

Purpose: Compare fare distributions between survivors and non-survivors

6. Survival by Embarkation Port

Chart Type: Bar Chart

X-Axis: Embarkation Port

Y-Axis: Survival Rate (%)

Purpose: Detect geographic boarding patterns

7. Family Size Impact

Chart Type: Line or Bar Chart

X-Axis: Family Size

Y-Axis: Survival Rate

Purpose: Identify optimal family size for survival